

Fermion Composition, Color Closure, and Bound-State Geometry

Volume 0. How fermionic systems compose once one fermion is no longer enough — rebuilt dependency-first, with the failures kept on the record.

About this book. Rebuilt as standalone R Theory from the former T5 essay-book, which was itself a "clean dependency-first reconstruction" of an earlier research draft. Each section states its inputs up front; no later section retroactively supplies an assumption an earlier result needed. The book's two architectures — **color** and **mass/binding** — are proved logically independent of each other.

Status key. **EXACT** proved mathematics · **CONDITIONAL** exact after a declared import or contract · **DECLARED** a mathematical extension, labeled as such · **IMPORT** accepted physics taken as input · **OBSTRUCTION / NEGATIVE** a proved or retained negative result.

Depends on: [Book 0](#) (composition rank, Clifford structure); [Book 20](#) (the half-angle witness); the internal radial Dirac–Coulomb paper (§5).

§1 Firewall

§2 Color

§3 QCD

§4 Two fermions

§5 Bound states

§6 Tests

§7 Failures

§8 E8

§9 Audit

§1 — The composition firewall

Before color, mass, or binding: what does it mean to compose two systems? For real finite-dimensional carriers there is an exact decomposition

$$\text{Sym}(V_A \otimes V_B) = [\text{Sym}(V_A) \otimes \text{Sym}(V_B)] \oplus [\text{Alt}(V_A) \otimes \text{Alt}(V_B)]$$

— the locally symmetric product span is *not* generically the whole composite symmetric sector. A globally symmetric sector ($\text{Alt} \otimes \text{Alt}$) can be invisible to any reconstruction restricted to local symmetric factors. Consequence, kept as a standing obstruction: **a list of locally visible subsystem coordinates cannot be**

assumed to determine the full composite state unless local tomography is separately established. No later success in this book may be used to erase this rank obstruction.

EXACT **Composition firewall.** New axioms: 0. New physics imports: 0. The color weight diagram is not the whole state of a hadron; pair coordinates are not the whole two-fermion state; a constituent list is not a complete interacting record.

§2 — Three states, roots, and $\mathfrak{su}(3)$: the correct logical order

Declare one mathematical extension: **DECLARED** $V_3 = \mathbb{C}^3$ — not derived from Book 0, declared. From it alone: the centered weights w_i form an equilateral triangle in the sum-zero plane; their pairwise differences form the A_2 root system (Cartan matrix $[[2,-1],[-1,2]]$, six roots, regular hexagon); and the commutator chain from one $\mathfrak{su}(2)$ plus one bridge closes to all of $\mathfrak{su}(3)$. One bridge is minimal *within the declared candidate class* — this is not a theorem that nature must realize color this way.

Then the demotion that fixes the logical order: the tetrahedral metric $G = \frac{1}{2}(I+J)$ restricts to a scalar multiple of the Euclidean metric on the A_2 sum-zero plane, so it is an **exact compatible realization** of the A_2 geometry — but A_2 does not uniquely select it and it does not uniquely select A_2 . The correct order is $\mathbb{C}^3 \rightarrow \mathbf{weights} \rightarrow A_2 \rightarrow \mathbf{su}(3)$, with tetrahedral geometry as an optional realization layer, never a foundation.

Three-state closure **EXACT**

A declared complex three-state carrier has a canonical weight triangle, the A_2 root system, and $\mathfrak{su}(3)$ Lie closure from one bridge. **OBSTRUCTION** Tetrahedral nonuniqueness: the tetrahedral metric is compatible, not foundational. **OBSTRUCTION** $\mathfrak{su}(3)$ -isomorphic mathematics does not derive physical QCD.

§3 — The QCD import: color map, singlets, and what is not derived

Now the physics enters by explicit import. **IMPORT** Standard QCD: quarks in the 3, antiquarks in the $\bar{3}$, gluons in the 8, $\text{su}(3)$ -valued connection — plus a **CONDITIONAL** projection contract identifying the abstract V_3 carrier with the QCD fundamental color carrier. After the import, the weight triangle becomes the red/green/blue Cartan weights ($30^\circ, 150^\circ, 270^\circ$), anticolors are antipodes, and the color/anticolor hexagon interlaces the root hexagon with an exact 30° offset. The standard singlet channels follow exactly: $3 \otimes \bar{3} = 1 \oplus 8$, $3 \otimes 3 \otimes 3 = 1 \oplus 8 \oplus 8 \oplus 10$, with the ϵ -contraction exactly color-invariant.

Two firewalls stay attached. First, the polar color map lives in the internal (T_3, T_8) Cartan plane — its angles are representation coordinates, **not** spatial directions; no diagram-shape resemblance licenses identifying color axes with x, y, z. Second, and crucially: **singlet representation theory does not derive confinement**. The tensors δ and ϵ prove invariant channels; they do not prove a linear potential, mass gap, asymptotic freedom, or the absence of colored asymptotic states. Confinement, g_s , Λ_{QCD} , and the Hamiltonian are imported dynamics.

CONDITIONAL **Triadic color closure.** Conditional on the QCD import, three fundamental color carriers possess an exact antisymmetric singlet channel — representation-theoretic closure, not dynamical confinement. **OBSTRUCTION** Neither confinement, nor couplings, nor a Hamiltonian follows from the Lie algebra.

§4 — Two fermions: the exact invariant and its independence

Independently of color — this path uses none of §§1–3 — bring two supplied masses into the half-angle coordinate language. With mass rapidity $\lambda_m = \ln(m_1/m_2)$, $q_m = \tanh(\lambda_m/2)$, relative rapidity η , and $q_v = \tanh(\eta/2)$, ordinary on-shell relativity gives the exact invariant

$$s = 2m_1m_2(\cosh \lambda_m + \cosh \eta)$$

equivalently $s/(m_1m_2) = [R(q_m)+R(q_m)^{-1}] + [R(q_v)+R(q_v)^{-1}]$ with $R(q) = (1+q)/(1-q)$: the same reciprocal transfer architecture appears twice inside one Lorentz scalar — mass asymmetry and relative motion. Two honesty guards are load-bearing: **q_m and q_v are independent physical data** (identifying them would be an extra ansatz, not a consequence), and **writing $m_1/m_2 = R(q_m)$ is a coordinate change, not a mass-generation mechanism** — it explains nothing about why the ratio has its observed value.

Two-fermion invariant EXACT

From supplied masses and ordinary special relativity — no color, no QCD, no tetrahedral geometry. q_m is not a Lorentz rapidity; repeated Cayley form does not reduce physical rank; no force, potential, or binding law appears; no particle mass is predicted.

§5 — Below threshold: the bound-mass coordinate and its conditional convergence

Continue the invariant below the free two-particle threshold on a declared analytic branch DECLARED

$\eta \rightarrow i\theta$, and set $u = \tan(\theta/2)$. Solving gives the exact bound-mass coordinate

$$u^2 = [(m_1+m_2)^2 - M^2] / [M^2 - (m_1-m_2)^2], \quad M^2 = (m_1-m_2)^2 + 4m_1m_2/(1+u^2)$$

— an invertible reparameterization of a supplied subthreshold mass, from which no force, spectrum, or binding dynamics follows. Weak binding gives $u^2 = B/(2\mu) + O(B^2)$, so the reduced mass appears as the leading conversion scale with no separate ansatz. The heavy-source limit gives $u^2 \rightarrow (m_2-E)/(m_2+E)$.

Then the book's strongest structural result. Import the standard point-Coulomb Dirac equation and the radial Dirac–Coulomb paper's circular-state theorem: on the $n_r = 0, \kappa < 0$ heavy-source branch, the radial spinor ratio is constant and $|G/F|^2 = (m-E)/(m+E)$ — the identical ratio. Hence, with no particle-mass ratio fitted,

$$u_{\text{pair}} = |G/F| = \text{sxp}(x)$$

exactly, on a declared state family, from two independent routes. The failure domain is kept visible: for $n_r > 0$ the ratio runs with radius (Prüfer phase plus independent amplitude), so **one global u cannot reconstruct a generic excited radial spinor.**

Independent coordinate convergence CONDITIONAL

Exact on the circular Dirac–Coulomb branch, conditional on imported Coulomb and Dirac–Coulomb physics. Not an empirical deviation from that physics. The $n_r > 0$ obstruction is part of the theorem's

statement, not a footnote.

§6 — Observational discipline: thresholds, transitions, and verified numbers

What actually determines u ? An absolute breakup threshold fixes it uniquely; a single transition frequency constrains only a relation between two absolute coordinates — and **increasing precision does not remove that rank deficiency**. The preferred control ladder is positronium $\rightarrow$ muonium $\rightarrow$ hydrogen $\rightarrow$ deuteron $\rightarrow$ heavier systems only with declared constituent masses. The numbers check (recomputed 2026-09-19 from the frozen inputs):

```
Hydrogen:  q_m  $\approx$  0.99891135885,  u_H(1S)  $\approx$  0.003648720323
            $\alpha/2 \approx$  0.003648676282  <  u_H  <  u_C  $\approx$  0.003648724858
           (u_C = infinite-source circular Dirac coordinate)
Deuteron:  B_d  $\approx$  2.22456637 MeV,  q_m  $\approx$  0.00068873505,
           u_d  $\approx$  0.0487186123  ( $\approx 13\times$  the atomic Coulomb scale)
Muonium:   q_m  $\approx$  0.99037 (leptonic, far from positronium's 0)
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EXACT **Deuteron control.** The deuteron sits near $q_m \approx 0$ while its u exceeds the atomic Coulomb scale by more than an order of magnitude — q_m and u are empirically independent, and no universal $u = \alpha/2$ law survives. **OBSTRUCTION** A difference from $\alpha/2$ or the infinite-source limit is not a Projection residual until the reference theory is matched at the same precision. **OBSTRUCTION** Scheme-undefined light-quark constituent masses do not define an invariant u .

Preregistration rule. Before testing any new mass law, record: constituent definition, bound-state definition, coordinate extraction, named reference theory, independent Projection prediction, parameter provenance, uncertainty propagation, invariant observable difference Δ_{op} , and failure criterion. Without these, the exercise is coordinate analysis, not empirical promotion. A residual may be read as a closure defect only when two independently specified theoretical routes produce the same observable — **measurement is not a second theory path**.

§7 — The frozen failure record

The original research program behind this book tried to find mass laws and failed. The failures are retained here as part of the book, not tuned away:

- **81/32 is not exact.** $R_\beta = (m_n - m_p)/m_e \approx 2.53098858276$ versus $81/32 = 2.53125$ — a residual of about -133.6 eV, many standard deviations from exact equality. It must not be presented as an exact mass formula.
- **Fifth-lattice proximity is not mass quantization.** Near-hits exist; negative lepton controls reject a universal all-masses-on-the-lattice rule, and same-isospin $d \rightarrow u$ tests reject the stronger claim.
- **Closure defects are not musical commas.** Coleman–Glashow and Gell-Mann–Okubo defects do not reproduce a clean Pythagorean or syntonic-comma law.
- **The order-five mass route failed.** An exact algebraic grading of chiral sectors does not generate observed fermion masses; restoring a Yukawa term needs imported scalar structure, and the grading then supplies no independent prediction. No new low-energy selection rule was promoted.

These negatives are inherited by any future flavor/mass program. They may not be silently reversed.

§8 — Optional appendix: the higher-carrier audit (firewalled)

The E8(−24) higher-carrier work is preserved here under its own dependency firewall: **no theorem in §§1–7 depends on this appendix; deleting it changes nothing above.** The honest negatives are kept: exact matrix particle/mirror exchange is not a physical oscillation law without dynamics; equal cyclic order is not identity of automorphism action; no fermion mass spectrum has been derived. The positive structural claims — the 248-direction reconstruction, the zero-residual particle/mirror matching, the $su(2,3) \oplus u(1)_Y$ Standard Model centralizer, the commuting SU(5) partner — are cited to the project's Verification Ledger, **not proved in this text**, and cannot be stated as established until that ledger and its code are rerun. They are recorded here as ledger-cited, nothing more.

§9 — Closure audit

Book 21 promotion audit

Projection-specific new equation of motion: none.

Independently determined coupling or mass parameter: none.

New invariant observable beyond invertible rewrites: none.

Quantitative deviation from a named reference theory: none predicted.

Therefore $\Delta_{\text{op}}(\text{Book 21}) = \emptyset$. Null hypothesis C0 retained.

Book 21 closure EXACT

Relative to the inherited foundations and the declared imports, this book establishes: the composition firewall; three-state $\rightarrow A_2 \rightarrow \text{su}(3)$ closure with the tetrahedral metric demoted to realization; conditional triadic color closure with confinement kept as imported dynamics; the exact two-fermion invariant with q_m, q_v independent; the below-threshold coordinate with its reduced-mass and heavy-source limits; the conditional Dirac–Coulomb convergence $u = |G/F| = \text{sxp}(x)$ on the circular branch; verified hydrogen/deuteron/muonium controls; the frozen failure record; and the firewalled higher-carrier audit. The color and mass architectures are logically independent — they do not derive one another. $\Delta_{\text{op}} = \emptyset$; C0 survives.

Dropped in this rebuild. The "closure reset" vocabulary (honestly labeled in the source but adding no dynamics — its own negative theorem says so) and the baryon–lepton "research lead" (an unpromoted hypothesis) are not carried into the main text. Open leads live outside the promoted ledger.

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A Volume 0 book of the R Theory rewrite. Source: the former T5 essay-book (v2 reconstruction), rebuilt standalone. Status discipline: exact / conditional / declared / import / obstruction — nothing promoted beyond its support. $\Delta_{\text{op}} = \emptyset$ throughout.